

HARSHITH SIDDARDHA MANNE

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SUMMARY

Cloud and DevOps Engineer with hands-on experience architecting production AWS infrastructure, automating deployments through Infrastructure as Code, and shipping containerized backend systems on ECS, EKS, and Kubernetes. Skilled in CI/CD automation, observability (Prometheus, Grafana, Loki, Datadog), and cost-efficient cloud-native design aligned with the AWS Well-Architected Framework. Extending these foundations into multi-tenant SaaS platforms and knowledge-graph-grounded AI systems.

EXPERIENCE

DEEP ALGORITHMS & SOLUTIONS

May 2023 - May 2024

Cloud/DevOps Engineer

- Architected a highly available 3-tier AWS infrastructure with Terraform and CloudFormation across VPC, ALB, ECS, RDS, and S3, cutting provisioning errors by 40% and infrastructure setup time by 35% through standardized IaC modules.
- Containerized Python-Flask microservices and deployed to ECS Fargate and Kubernetes (EKS) with Helm-managed workloads, Horizontal Pod Autoscaling, and Nginx ingress, supporting 2x traffic growth without performance degradation.
- Engineered CI/CD pipelines with AWS CodePipeline, CodeBuild, and CodeCommit, Jenkins, and GitHub Actions following GitOps workflows, reducing deployment failures by 30% and accelerating release cycles by 40% across dev and staging environments.
- Built observability with Prometheus, Grafana, and Loki for cluster metrics, application logs, and SLO dashboards; integrated CloudWatch alarms and Datadog-style golden-signal monitoring to cut incident MTTR by 35%.
- Optimized Docker multi-stage builds and ECS task definitions to cut deployment time by 45%; configured autoscaling, Multi-AZ RDS, and developed Lambda-based ML batch pipelines in private subnets using SSM, Secrets Manager, S3, and NAT Gateway for secure outbound access.

BACKFLIPT (XENOVIOUS, INC.)

Jan 2023 - Apr 2023

Associate Software Developer Intern

- Designed high-concurrency RESTful services in Java, Spring Boot, and Reactive Spring for the International Data Corporation (IDC) project, improving API response efficiency by 30% under load.
- Integrated backend APIs with React and React Native clients using Redux state management, reducing frontend-backend integration defects by 25%.
- Tuned and resolved production issues across AWS-hosted environments, cutting incident turnaround time by 35% through faster debugging, monitoring, and release support.

DEEP ALGORITHMS & SOLUTIONS

Jan 2021 - Mar 2022

DevOps & Backend Engineer Intern

- Engineered RESTful APIs for an AI NLP chatbot using Python, Flask, and Django with JWT authentication, request validation, and SQL/NoSQL persistence; containerized services with Docker and Docker Compose.
- Provisioned AWS infrastructure using CloudFormation across EC2, S3, IAM, and ALB with autoscaling and lifecycle policies; integrated CloudWatch and CloudTrail for monitoring and audit visibility.
- Built end-to-end CI/CD pipelines with AWS CodePipeline, CodeBuild, and CodeCommit using PR-based branching across dev, stage, and prod environments.

PROJECTS

Infratrix — Dual Knowledge Graph for AWS Cost Optimization

Aithrex

- Building Infratrix (under Aithrex), a multi-tenant SaaS platform that ingests live AWS topology, CloudWatch metrics, and Cost Explorer data, projecting resources onto a 12k-node / 14k-edge knowledge graph using NetworkX MultiDiGraph.
- Modeling AWS services, 104 optimization patterns, and cost factors to surface ranked recommendations through a deterministic 5-stage solver, validated end-to-end with a scenario-based regression harness.
- Implemented graph-grounded RAG chat on Amazon Bedrock with Claude Sonnet, citation-required prompts, and a heuristic-then-upgrade polling flow to operate within API Gateway's 30-second HTTP limit.
- Built a live AWS scanner using STS AssumeRole and boto3 across 12+ services including EC2, RDS, EKS, Lambda, DynamoDB, and S3, feeding both the solver and a hierarchical SVG architecture view.
- Implemented multi-tenant RBAC with Postgres row-level isolation, JWT and refresh tokens, and compliance filtering for SOC 2, HIPAA, and PCI-DSS; deployed FastAPI as containerized Lambdas via Terraform with GitHub Actions CI/CD.

TECHNICAL SKILLS

Cloud / DevOps / IaC: AWS (EC2, ECS, EKS, Lambda, S3, RDS, DynamoDB, VPC, IAM, ALB, API Gateway, CloudFront, KMS, Bedrock), Linux, Terraform, CloudFormation, Docker, Kubernetes / EKS (Helm, HPA), Nginx, CI/CD (GitHub Actions, AWS CodePipeline / CodeBuild), Site Reliability Engineering (SRE)

Observability: Prometheus, Grafana, Loki, Datadog, CloudWatch (Logs, Metrics, Insights, Alarms), CloudTrail, Distributed Tracing, Golden Signal Monitoring

Backend / AI Systems: Python (FastAPI, Flask, Django, Asyncio, SQLAlchemy), Java (Spring Boot, Reactive Spring), TypeScript, REST APIs, Microservices, JWT / OAuth / RBAC

Data / AI / Knowledge Systems: PostgreSQL, MongoDB, Alembic, Amazon Bedrock, Anthropic SDK, RAG Pipelines, Agentic AI Workflows, Knowledge Graphs (NetworkX), MCP, FinOps, AWS Well-Architected Framework

OS / Networking / Scripting: Linux (Ubuntu, Amazon Linux), Bash, Shell Scripting, TCP/IP, DNS, VPC Subnetting, Security Groups, NACLs, NAT Gateway, Load Balancing

EDUCATION

Arizona State University, Tempe, AZ | Master of Science in Information Technology

May 2026

Jawaharlal Nehru Technological University (JNTU) Kakinada | Bachelor's in Computer Science

May 2023